TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Lâm Hữu Khánh Phương		phuonglhk@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Lê Nguyễn Quang Đức	SE192066	0901430379	quangducle.127@gmail.com	Leader
2	Vũ Anh Tuấn	SE190510	03757FA26SE183	tuanmeh123@gmail.com	Member
3	Trần Tiến Đạt	SE180773	0339537689	dattt1608@gmail.com	Member
4	Phạm Viết Lợi	SE181843	0964568267	loipvse181843@fpt.edu.vn	Member

Record of Changes

Version	Date	A/M/D	In Charge	Change Description	Reference
1.0	[verify]	A	Academic Supervisor	Original approved registration; unchanged title, code and team metadata.	Official project assignment decision [attach]
1.1	[verify]	M	Team Leader	Post-Review 1 refocus to STEM education; clarified SE platform versus AI workstream and bounded	Review 1 feedback / supervisor minutes [attach]

					physical demonstration.	
1.2	10/10/2026	M		Team Leader	Expanded actor-specific requirements, organization/resource governance, experiment lifecycle, measurable quality targets and work-package acceptance.	Scope reconciliation against v1.1; supervisor confirmation pending
1.2	10/10/2026	A		Team Leader	Added existing-systems gap, evidence assumptions, preliminary exclusions, risks and fallbacks.	Guideline v1.2; review minutes reference pending

* A = Added; M = Modified; D = Deleted. Dates and references marked for verification must not be represented as approved decisions.

3. Register content of Capstone Project FA26SE183

(*) 3.1. Capstone Project name:

3.1.1. English: ***VISTA: Vision-Language-Action System for Tabletop Manipulation with Dual Robot Arms***

3.1.2. Vietnamese: ***VISTA: Hệ thống Thị giác - Ngôn ngữ - Hành động điều khiển hai cánh tay robot thao tác trên bàn***

Abbreviation: VISTA

3.2. Main proposal content (including result and product)

a) Context:

Robotics and Artificial Intelligence (AI) are increasingly important in STEM education, but many learning experiences remain limited to screen-based coding or demonstrations that do not connect programming, data, AI models, and physical robot behavior in one learning workflow. Low-cost tabletop robot arms and open-source robot-learning frameworks make hands-on activities more accessible, but learners and instructors still need a structured environment to organize lessons, assignments, robot practice, experiment records, and learning outcomes.

VISTA addresses this educational and software-integration need by combining a Robotics & AI Learning Platform with a Vision-Language-Action (VLA) robotic system. The platform is the primary

software product for the Software Engineering team; the robot is the physical practice device integrated through a local Robot Runtime. The project will demonstrate a bounded educational workflow rather than attempt to deliver a production-ready learning management system or a general-purpose autonomous robot.
The post-Review 1 scope is focused on STEM education. Disability-assistance and household-assistance scenarios are not part of the current project scope. The registered project title is retained, while the proposed solution, requirements, deliverables, and evaluation are refined to align with the review feedback.
- Existing approaches and gap: General learning-management systems can distribute assignments and grades, while robot-learning toolkits such as Hugging Face LeRobot support robot data and policy workflows. Neither is assumed to provide, out of the box, the complete VISTA class-to-physical-experiment-to-feedback workflow. This gap will be verified through a feature comparison and stakeholder review; it is not a claim that no comparable platform exists.
- Evidence and assumptions: Availability of the dual-arm tabletop configuration, supported cameras, and AI-service endpoints depends on hardware inventory and the AI workstream. The team will document the verified configuration, the baseline workflow and measured outcomes rather than invent market statistics or claim unverified robot autonomy.
References: Hugging Face LeRobot documentation; the confirmed robot hardware documentation; supervisor Review 1 feedback (formal meeting record to be attached).

b) Proposed Solutions:

VISTA is a role-governed Robotics & AI Learning Platform integrated with a local tabletop robot practice environment and an AI/model service boundary. Its core value is a traceable teaching workflow: an administrator provisions users and resources; a teacher prepares a lesson and assigns a robot activity; a student prepares and submits an approved program or model configuration; the local runtime executes the bounded task after safety checks; the platform records evidence and the teacher evaluates the result. The web platform owns educational and administrative data; the local runtime owns device communication and real-time safety; AI model development is provided by the AI workstream.
- Learning Platform (SE core): provide basic user roles and access control, class/course/lesson and assignment workflows, learning activity records, experiment history, student progress, and teacher review.
- Robot Programming & Practice: provide a representative beginner-friendly programming activity, such as block-based sequencing for pick-and-place, and connect the activity to the local Robot Runtime for physical practice where hardware is available.
- AI/Data Workflow Integration: represent dataset/model versions, training or evaluation job status, model eligibility, and deployment requests through defined interfaces. The AI team owns VLA model development, training/fine-tuning, inference, and AI-specific evaluation; the SE team integrates these capabilities into platform workflows.
- Local Robot Runtime: connect the platform to the robot and cameras, receive an authorized experiment/model configuration, execute the approved task, report status and telemetry, and stop safely when communication or execution conditions become unsafe.
- Teacher/Admin Portal: manage classes, lessons, assignments, device/resource allocation, review experiment evidence, and monitor learning progress.
- Integration & Observability: define APIs/contracts between the platform, AI services, and Robot Runtime; maintain experiment records, error states, and audit-friendly logs for the end-to-end workflow.

The representative end-to-end activity will be a constrained tabletop object-sorting or pick-and-place task. Advanced bimanual manipulation, broad policy benchmarking, a complete simulation platform, and a production-scale multi-school service are not guaranteed deliverables.
- Scope decision (post-Review 1): RETAIN the STEM learning platform, guided tabletop practice, experiment evidence and teacher review; NARROW robot control to one bounded pick-and-place/object-sorting activity, AI integration to agreed model/dataset metadata and callable service interfaces, and reporting to basic learning progress and experiment history. REMOVE disability/household assistance, general-purpose autonomous manipulation, large-scale model benchmarking and a production-grade LMS. These decisions are inferred from the approved v1.1 text; the team must map each to its Review 1 minutes before sign-off.

c) Functional Requirements:

- Student: view assigned courses/lessons; open an assigned activity; use the supported robot-programming interface; submit/run an experiment when the required robot/runtime is available; view their experiment history and feedback.
- Teacher: create or manage lessons and assignments for assigned classes; allocate an activity and eligible robot/resource; inspect submitted experiment records; provide an assessment or feedback; review basic class/student progress.
- School/System Administrator: manage basic accounts, roles, class membership, and resource/device registration; grant access to classes and resources according to defined permissions.
- Learning workflow: enforce that a student can access only assigned or otherwise authorized learning resources; record the lesson/activity, student or team, program/configuration, execution status, timestamp, and result for each experiment.
- AI/model workflow integration: display dataset/model metadata and training/evaluation status returned by the AI component; associate a model version with its dataset/job/evaluation metadata when provided; allow deployment requests only for a model that meets the agreed compatibility/eligibility rules.
- Robot Runtime integration: request an experiment run through a defined interface; display runtime connection/execution status; record success, failure, or interruption; prevent a new control session from conflicting with an active session for the same robot.
- Safety and failure handling: stop or reject execution on loss of required communication, invalid/stale inference output, or a runtime-reported safety fault; show an actionable error state instead of silently continuing.
- Reporting: provide basic experiment history and progress views for students and teachers. Detailed analytics and institution-wide reporting are optional extensions.
- All authenticated users: sign in and sign out; view and update permitted profile information; view their organization and role context; receive meaningful access-denied and session-status messages; view only records authorized for their membership.
- Student — learning: view enrolled classes and published lessons, read activity instructions and prerequisites, open assigned programming tasks, save supported action sequences/configurations, validate submissions, and review teacher feedback.
- Student — robot practice: request an available eligible robot; view connection and readiness; explicitly start or cancel an authorized run; observe current state and outcome; inspect own session history, errors and experiment evidence. Students cannot override safety limits or take over another active session.
- Teacher — curriculum: create/edit courses and lessons within assigned classes, publish activities, set due dates and expected outcomes, assign students or teams, and choose an eligible device/model from the available resource pool.

- Teacher — review: inspect submissions, execution status and evidence, record comments or assessments, request a retry, and view basic per-student and per-class completion summaries; teachers cannot grant system-wide resources beyond their delegated rights.
- School/System Administrator — identity: create, activate, deactivate and assign users to organizations/classes; maintain role mappings and resource entitlements; register devices and update availability or maintenance status.
- School/System Administrator — governance: manage model eligibility metadata, configure approved safety profiles in coordination with the robotics team, inspect audit/session records and resource utilization, and revoke access that must prevent new sessions.
- Platform — authorization: verify authentication, organization membership, role, class assignment, resource grant, model compatibility and session conflict before accepting an experiment; return explicit reasons when a check fails.
- Platform — experiment tracking: generate a unique experiment/session identifier; associate learner, class, lesson, activity, robot, configuration and optional model version; persist state transitions, timestamps, outcome and available evidence; expose filtered history.
- Local Robot Runtime (integrated component, not a human actor): establish a local robot/camera connection, validate command format and safety limits, report readiness and telemetry, execute bounded approved tasks, implement stop/abort, and return a final result or explicit failure reason.
- AI service (integrated component, not a human actor): expose agreed dataset/model/job metadata and supported inference/deployment operations; return version and status information for traceability; surface timeout, incompatible model and unavailable-service states without falsely marking the experiment successful.
- Exception flows: unavailable robot, expired permission, incompatible model, disconnected USB/camera, stale AI output, execution timeout, emergency stop and repeated run requests must lead to rejected/interrupted/failed states with an understandable UI message and retained audit evidence.
- Cross-role and integration rules: A robot must have at most one active authorized execution session; an expired allocation cannot initiate motion; every command is validated against supported actions and workspace limits; a failed or interrupted session remains reviewable; students cannot access other classes' records. AI job statuses are displayed only when the connected AI service exposes them; no in-platform model training is promised.

d) Non-Functional Requirements:

- Safety: real-robot execution must be initiated explicitly, use agreed workspace/joint limits and emergency-stop procedures, and fail safely when required communication or runtime checks fail. Hardware safety thresholds must be validated with the robotics team before live operation.
- Reliability: platform, AI service, and Robot Runtime failures must be represented as explicit states (for example, queued/running/completed/failed or disconnected) and must not be reported as successful experiments when no success result is received.
- Usability: a student should be able to complete the representative lesson through the documented UI flow without editing backend data manually; teacher workflows should expose assignment and assessment status clearly.
- Traceability: each experiment record should identify the relevant learner/team, activity, program or configuration, model version where applicable, robot/runtime, time, status, and available result evidence.

- Security and privacy: enforce role- and class-based authorization for protected resources; do not expose credentials or tokens in client-visible logs; only store the learner information required for the capstone demonstration.
- Maintainability: use documented API contracts between the Learning Platform, AI component, and Robot Runtime; keep integration errors observable and test the critical business rules.
- Performance: establish measurable response-time targets for platform operations and runtime commands during implementation testing. Real-time robot control and AI inference performance are owned and validated with the robotics/AI components; the platform must report delays/failures rather than claim a fixed control frequency before measurement.
- Portability: the platform should communicate with a local Robot Runtime through a documented interface so the learning workflow is not tightly coupled to one laptop or camera implementation.
- Measurable authorization: 100% of predefined negative permission tests must reject unauthorized access to classes, robot assignments and other users' experiment records; the backend enforces access independently of the UI.
- Session exclusivity and consistency: At most one active control session is permitted per physical robot. Duplicate requests must not create parallel execution; terminal status must reflect the runtime acknowledgment rather than an optimistic UI assumption.
- Proposed responsiveness: Under the agreed demonstration load, target p95 <= 3 seconds for ordinary authenticated page/API operations, excluding model inference and physical execution. This is a provisional acceptance target to validate with the supervisor and measurement setup.
- Proposed trace completeness: At least 95% of executed demonstration attempts should have a persisted record linking actor, activity, robot, start/end times and terminal status; any missing evidence must be reported rather than fabricated.
- Fault tolerance: When the AI endpoint is unavailable, manual/guided learning and history browsing remain usable; the platform shows an explicit unavailable state. If the runtime loses required communication during execution, the local safety policy must halt/reject commands and report interruption.
- Privacy and data handling: passwords are hashed, secrets are kept server-side, sensitive logs are redacted, and student records are limited to data necessary for the capstone. Access to class records is scoped to membership and role.
- Usability evaluation: A documented scenario-based usability test will measure task success for creating an assignment, completing a student experiment and reviewing feedback; an initial target of >= 80% task completion is proposed, subject to supervisor approval.
- Deployability and reproducibility: provide environment configuration, migration/setup steps, API examples and a reproducible demonstration dataset; critical authorization and session-state rules receive automated tests.
- Preliminary acceptance criteria (subject to hardware and supervisor confirmation): 95% of ordinary authenticated page/API requests complete within 3 seconds in the agreed demonstration environment; each accepted experiment command receives an acknowledgement or explicit timeout within 5 seconds; all tested unauthorized role/class actions are denied; every executed run records actor, lesson/activity, robot, model/configuration where applicable, timestamps, final state and outcome; emergency stop and loss-of-communication behavior are verified in supervised tests. These are proposed targets, not verified measurements.

e) Theory & Practical:

Theory

- STEM and computational-thinking pedagogy: sequence, condition, loop, debugging, and scaffolded learning activities.
- Robot programming and manipulation: basic action sequencing, robot state, camera-based observation, and safe execution on a tabletop robot.
- AI/data lifecycle: dataset collection and labeling, training/evaluation job concepts, model versions, deployment, and experiment-based evaluation.
- Software Engineering: requirements analysis, role-based authorization, workflow/state modeling, API integration, logging, testing, and traceability.
- Technical references to validate: Hugging Face LeRobot documentation for dataset/policy integration; official hardware/runtime manuals for robot limits and emergency stop; OWASP ASVS for authentication/authorization; ISO/IEC 25010 as a software-quality reference. Exact versions and applicable requirements will be cited in Report 2.
- Preliminary references for design and verification: ISO/IEC/IEEE 29148:2018 (requirements engineering); ISO/IEC 25010:2023 (software product quality); NIST SP 800-53 Rev. 5 (access control and audit principles); ROS 2 documentation (robot middleware and communication concepts); Hugging Face LeRobot documentation (robot learning data/model integration). Applicability to the selected hardware and API stack will be checked during architecture design.

Practical

- Develop the Learning Platform workflows for a representative class, lesson, assignment, robot activity, experiment, and teacher assessment.
- Integrate a supported block-based or guided programming activity with the local Robot Runtime for a bounded physical task.
- Integrate AI-team-provided model/data/training/evaluation capabilities through agreed APIs or service contracts; do not duplicate the AI team's model-development responsibility.
- Demonstrate the end-to-end workflow on available tabletop robot hardware, with simulation or a recorded/mock service used only where necessary and clearly identified in the report.
- Test key permissions, state transitions, failure handling, experiment traceability, and the safe-stop behavior of the integrated system.
- Validate the actual robot/camera/USB setup, record the safety checklist and run bounded experiments with known objects and a documented tabletop workspace.
- Conduct a requirements-to-test traceability review, negative permission tests, duplicate-session tests, AI-service timeout tests, and safe-stop drills before the final physical demonstration.

f) Products (Expected Deliverables):

- VISTA Learning Platform: student-facing learning activity and experiment workflows, with basic teacher/admin functions.
- Course/Lesson/Assignment Demonstration: at least one representative Robotics & AI learning activity and its assignment/review flow.
- Robot Programming Interface: a bounded beginner-friendly programming experience connected to the robot practice workflow.
- Local Robot Runtime Integration: documented interface and demonstrable connection/execution/status reporting for the supported robot setup, subject to hardware validation.
- AI/Data Workflow Integration: model/dataset/job metadata and deployment workflow integrated with the AI component; the exact available functions depend on the agreed AI API contract.

- Experiment Record & Progress Views: stored experiment outcomes and basic student/teacher review.
- Engineering Documentation: system architecture, requirements and business rules, API contracts, data model/ERD, workflow/state diagrams, test evidence, deployment/setup guide, and known limitations.
- Final Demonstration and Evaluation Report: evidence from the representative end-to-end activity, including recorded successes/failures and limitations. The project will not claim a policy benchmark unless that experiment is actually completed.
- Organization and Resource Administration: organization/class membership, role grants, device registration, resource assignment and audit views necessary to run the teaching workflow.
- Validation and Evidence Package: representative seeded courses, assignments, students and experiments; test cases for authorization, safety/failure states, and measured end-to-end demonstration outcomes.
- Scope-to-deliverable check: All deliverables above serve the retained or narrowed pillars. No disability/household-assistance application, general robot policy benchmark suite, commercial LMS, or independent AI foundation-model training product is included. Any new deliverable requires a corresponding scope decision and supervisor review.

g) Proposed Tasks:

Team of 4 members. Each member leads one work package with shared responsibility for integration, documentation, verification and the final defense. Scope is divided by accountable deliverables rather than by unverified advanced robotics features.
- WP1 – Requirements, Architecture, Governance & QA (Project Lead / BA / Backend): confirm the scope and stakeholder scenarios; preserve approved project identity; define roles, organization/class/resource rules, ERD, architecture, API contracts and session state machine; build authentication and core administrative backend; coordinate integration and own system-level test plan. Done when: core permission rules and class/resource administration pass agreed tests, architecture/contracts are documented, and traceability links requirements to evidence.
- WP2 – Local Robot Runtime & Safety (Robotics Integration Developer): configure supported robot arms, USB/cameras and calibration; implement local connection, readiness, approved task execution, state/telemetry reporting, emergency stop and failure handling; validate one constrained pick-and-place or sorting demonstration. Done when: the runtime accepts authorized commands, prevents conflicting execution, reports outcome and demonstrates safe interruption on the available hardware.
- WP3 – AI/Data Service & Experiment Integration (Integration Developer): define model/dataset/job metadata contracts with the AI workstream; implement model eligibility/deployment-request workflow, service adapters, timeout/error handling, session and experiment evidence persistence; coordinate model-version traceability. Done when: the platform can display available model metadata, reject incompatible requests and link a supported AI-assisted run to its model and experiment record; model training is not claimed as SE ownership.
- WP4 – Learning Experience, Full-stack UI & Evaluation (Frontend / Full-stack Developer): implement student, teacher and administrator interfaces; courses, lessons, assignments, guided programming, experiment start/status/history, feedback and progress; conduct end-to-end usability checks and prepare the demo. Done when: teacher and student can

complete the full lesson–experiment–review path through the UI without manual database changes.
All members participate in integration, validation of business rules, documentation, and preparation of the final demonstration. Work-package allocation may be adjusted to match the team's confirmed assignment while preserving clear ownership.
- Cross-package acceptance: The integrated demonstration must show an administrator granting a resource, a teacher assigning an activity, a student executing an authorized bounded experiment, a recorded outcome, teacher feedback and at least one visible failure/safety case.
- Work-package coordination: WP1 owns learning/domain requirements, authorization and backend contracts; WP2 owns safe local execution and robot interfaces; WP3 owns AI/data service adapters and evaluation metadata; WP4 owns guided learning UI, assessment and user workflows. Integration tests cover a full Teacher → Student → Robot Runtime/AI Service → Teacher feedback cycle. Individual owners must be confirmed by the team rather than assumed.

3.3. Research Information

a. Research Problem / Research Question

How can a software learning platform integrate guided robot-programming activities, AI/model workflow services, and a local tabletop robot runtime into a traceable STEM learning and experiment workflow for students and teachers?

b. Research Objectives

- Define and implement the core Learning Platform workflows for classes, lessons, assignments, robot activities, experiments, and teacher review.
- Integrate the platform with a local robot runtime and AI/data services through documented interfaces and explicit state transitions.
- Demonstrate one bounded Robotics & AI learning activity on the supported robot setup, and record experiment outcomes for review.
- Evaluate the integrated workflow against defined functional, authorization, traceability, usability, and failure-handling criteria.

c. Research Scope & Methodology

The project uses an iterative Software Engineering approach: stakeholder and requirement analysis; architecture and data/workflow modeling; incremental implementation; API integration; and verification through scenario-based tests and an end-to-end demonstration. Evaluation will focus on whether authorized users can complete the representative learning workflow, whether experiment records are traceable, and whether key error and safety states are handled as specified. AI model accuracy and policy-level benchmarking are evaluated only to the extent agreed with and evidenced by the AI workstream.
In scope: a representative STEM learning pathway, core Learning Platform workflows, one bounded robot-programming/physical-practice activity, AI/model workflow integration, experiment records, basic teacher assessment, and essential safety/failure handling.
Out of scope for the current capstone scope: disability or household assistance; a full commercial LMS; a standalone mobile application or VR experience; broad multi-policy benchmarking (including ACT/Diffusion Policy comparisons unless separately approved); building a new foundation/VLA model from scratch; unrestricted autonomous manipulation; and production-scale operation across many schools.

d. Expected Scientific / Engineering Contribution

The expected contribution is an integrated, testable software workflow that connects Robotics & AI learning activities with a physical robot practice environment, while preserving role-based access, experiment traceability, explicit service states, and teacher feedback. The project will document the integration boundary between the Learning Platform, AI services, and local Robot Runtime so the workflow can be extended in later work.

e. Related Works / Literature Review (Preliminary)

- Hugging Face LeRobot: open-source robot learning, teleoperation, dataset conventions, and policy integration.
- SO-101 tabletop robot ecosystem: low-cost robot hardware and practical robot-learning activities.
- STEM and block-based programming environments: scaffolded computational-thinking activities for learners.
- Vision-Language-Action (VLA) research and the model/service selected by the AI workstream: used as an integrated robot capability rather than a promise that the SE workstream will develop a new VLA architecture.
- Software Engineering references for requirements engineering, role-based access control, workflow/state modeling, API design, and system testing.
- Preliminary evidence plan: Compare VISTA against a generic LMS and a robot-learning toolkit using a documented feature matrix; measure workflow completion, permission enforcement, session traceability and error handling on the demonstrated configuration. No generalizable claim about VLA accuracy will be made without AI-team evidence.
- Existing systems and gap (preliminary): Conventional LMS products support learning content and submissions but do not themselves provide bounded robot-runtime authorization and per-experiment robot evidence. Robotics toolkits support robot control or dataset/model workflows but do not, by themselves, provide classroom assignment and teacher-review processes. VISTA addresses this integration gap for a single supervised STEM activity; this is an architectural comparison, not a claim of unique market novelty.
- Risks and contingencies: (R1) Hardware/camera unavailability → preserve the same API contract and demonstrate using simulation or recorded traces, clearly labeled as fallback. (R2) AI service/model not ready → use a contract-compliant stub and demonstrate the integration states without claiming trained-model performance. (R3) Unsafe or stale robot commands → validate limits locally, require operator supervision and provide emergency stop. (R4) Network interruption → terminate/mark session interrupted and preserve partial evidence. (R5) Scope growth → require change log, product cross-check and supervisor approval before adding deliverables.
- Preliminary out of scope (with rationale): household/disability assistance (removed use case); unrestricted dual-arm autonomous manipulation (safety and feasibility); general-purpose model training/benchmark platform (separate AI workstream); commercial-grade LMS features such as tuition/payment or multi-school billing (not needed for the teaching demonstration); unsupervised remote operation of physical robots (safety); unlimited multi-robot orchestration (not required for the bounded scenario).
- Change governance: Version dates, Review 1 decision IDs and approval references must be copied from actual meeting minutes; do not treat placeholders as evidence. No change to the official project code, bilingual titles, abbreviation, supervisor or members is authorized.

Any change beyond clarification/narrowing must be listed separately and escalated to the supervisor/faculty.

4. Other comments (propose all relative things if have):

Post-Review 1 revision: the approved project code/title and group/supervisor information remain unchanged. The SE contribution is a STEM Robotics & AI Learning Platform with governed learning workflows, local runtime integration and traceable experiments. The revised scope retains the educational platform, guided programming, resource governance, safety and AI service integration, while narrowing physical demonstrations to a bounded tabletop task. Any further scope change beyond the approved adjustment requires supervisor review.
Out of scope (preliminary): disability/household assistance, commercial billing/licensing, a complete production LMS, standalone mobile or VR control, internet-based remote real-time robot control, unrestricted autonomous manipulation, guaranteed advanced bimanual coordination, new foundation-model training, and production-scale multi-school deployment.
Risks and fallback: (1) hardware delay -> mock runtime for platform testing, clearly labeled, plus physical demo when available; (2) AI API delay -> versioned mock responses and manual activity path; (3) safety/calibration uncertainty -> restrict workspace and require robotics sign-off; (4) schedule pressure -> prioritize the single end-to-end teaching workflow.
Pending team verification with supervisor: meeting minutes/change references, robot configuration, AI API contract, performance thresholds and final acceptance criteria.

.....
.....
.....
.....
.....
.....

HCM, date 19/08/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*